

Reply

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6. Crystal E, Garfinkle MS, Conolly SS, Ginger TT, Sleik K, Yusuf SS. Interventions for preventing post-operative atrial fibrillation in patients undergoing heart surgery. *Cochrane Database Syst Rev* 2004;4: CD003611.
7. Mozaffarian D, Psaty BM, Rimm EB, et al. Fish intake and risk of incident atrial fibrillation. *Circulation* 2004;110:368–73.
8. Brouwer IA, Heeringa J, Geleijnse JM, Zock PL, Witteman JC. Intake of very long-chain n-3 fatty acids from fish and incidence of atrial fibrillation. The Rotterdam Study. *Am Heart J* 2006;151: 857–62.
9. Frost L, Vestergaard P. n-3 fatty acids consumed from fish and risk of atrial fibrillation or flutter: the Danish Diet, Cancer, and Health Study. *Am J Clin Nutr* 2005;81:504.
10. León H, Shibata MC, Sivakumaran S, Dorgan M, Chatterley T, Tsuyuki RT. Effect of fish oil on arrhythmias and mortality: systematic review. *BMJ* 2008;337:a2931.
11. Brouwer IA, Raitt MH, Dullemeijer C, et al. Effect of fish oil on ventricular tachyarrhythmia in three studies in patients with implantable cardioverter defibrillators. *Eur Heart J* 2009;30:8206.

Reply

We thank Drs. Saravanan and Davidson for their insightful comments regarding details of our omega-3 polyunsaturated fatty acid (ω -3 PUFA) review and appreciate the opportunity to clarify issues that might have been prone to misinterpretation (1). We did not imply that the DART (Diet and Reinfarction) study (2), GISSI Prevenzione (Gruppo Italiano per lo Studio della Sopravvivenza nell' Infarto Miocardico-Prevenzione) study (3) and JELIS study (Japan EPA Lipid Intervention Study) (4) were trials with a uniform population but in fact stated that they represent an aggregate of evidence toward both secondary and primary prevention. They correctly point out that the DART study (2) and the GISSI-Prevenzione study (3) are secondary prevention studies with a significant ω -3 PUFA benefit in over 13,000 post-myocardial infarction patients. However, even in secondary prevention, not all studies have demonstrated benefit (1). The JELIS trial (4), in contrast, tested the effects of eicosapentaenoic acid (EPA) in mostly a primary prevention cohort ($n = 14,981$ with hypercholesterolemia treated with statins). Here, the effect size was essentially identical (18% vs. 19%) to that seen in the secondary prevention group ($n = 3,664$), but the 5-year event rate was nearly 6-fold higher in the secondary prevention group (12% vs. < 2%). As Drs. Saravanan and Davidson correctly point out, in the entire primary prevention group, the benefits of EPA were not quite statistically significant (hazard ratio: 0.82; 95% confidence interval: 0.63 to 1.06; $p = 0.13$). In a subgroup of JELIS primary prevention patients (5), those with triglycerides ≥ 150 mg/dl and high-density lipoprotein cholesterol <40 mg/dl had event rates that were nearly 2-fold higher than those without this lipid pattern. In patients with this high-risk lipid combination, EPA reduced major cardiac event rates by 53% ($p = 0.043$). Along with the notable epidemiological data, we believe that the overall evidence suggests benefits of ω -3 PUFA at least in high-risk risk primary prevention patients as well as in those needing secondary prevention of coronary heart disease.

We agree with Drs. Saravanan and Davidson that our single, brief paragraph on the impact of ω -3 PUFA in atrial fibrillation (AF) was oversimplified (in an effort to make a lengthy review more concise). Although the observational study by Mozaffarian et al. (6) suggested that high fish intake was associated with a 30% reduction in AF over 12 years, the Rotterdam Study (7) found no such effect. Only 1 small randomized controlled study of 160 post-bypass patients was performed to assess the benefits of ω -3

PUFA on the development of postoperative AF (8). The results of this small study were remarkable in support of ω -3 PUFA, including 18.1% absolute risk reduction and 54.4% relative risk reduction (or only 5.5 patients needed to be treated to prevent 1 episode of AF). As this study points out and as we have discussed elsewhere in more detail (9), these results compare favorably to the results of another large meta-analysis (58 studies including 8,565 participants) (10), which examined the effects of amiodarone, sotalol, and beta blockers on post-bypass AF. In comparison, the results with ω -3 PUFA post-bypass seem to be similar or even superior to these other treatments to prevent AF. Clearly, larger studies are needed and some are underway (11) to assess the impact of ω -3 PUFA in primary and secondary prevention of AF. Additionally, higher doses of ω -3 PUFA (e.g., 2 to 5 g/day) need to be studied in various cardiovascular diseases (CVD), including AF, and studies are needed to determine the relative benefits of EPA and docosahexaenoic acid (DHA) in CVD. It also is not known what impact DHA or EPA alone has in disease modification, these areas being the subject of intense ongoing investigation.

Finally, we agree with Drs. Saravanan and Davidson that one could debate the details of the strengths and weaknesses of each study. However, we believe the totality of the evidence suggests an overall beneficial effect of ω -3 PUFA for CVD protection, a view that we are not alone in professing (12–14).

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REFERENCES

1. Lavie CJ, Milani RV, Mehra MR, Ventura HO. Omega-3 fatty acids polyunsaturated fatty acids and cardiovascular diseases. *J Am Coll Cardiol* 2009;54:585–94.
2. Burr ML, Fehily AM, Gilbert JF, et al. Effects of changes in fat, fish, and fibre intakes on death and myocardial reinfarction: Diet And Reinfarction Trial (DART). *Lancet* 1989;2:757–61.
3. Dietary supplementation with n-3 polyunsaturated fatty acids and vitamin E after myocardial infarction: results of the GISSI-Prevenzione trial Gruppo Italiano per lo Studio della Sopravvivenza nell'Infarto miocardico (errata in *Lancet* 2001;357:642 and *Lancet* 2007;369:106). *Lancet* 1999;354:447–55.
4. Yokoyama M, Origasa H, Matsuzaki M, et al. Effects of eicosapentaenoic acid on major coronary events in hypercholesterolaemic patients (JELIS): a randomised open-label, blinded endpoint analysis (correction in *Lancet* 2007;370:220). *Lancet* 2007;369:1090–8.
5. Saito Y, Yokoyama M, Origasa H, et al. Effects of EPA on coronary artery disease in hypercholesterolemic patients with multiple risk factors: sub-analysis of primary prevention cases from the Japan EPA Lipid Intervention Study (JELIS). *Atherosclerosis* 2008;200: 135–40.
6. Mozaffarian D, Psaty BM, Rimm EB, et al. Fish intake and risk of incident atrial fibrillation. *Circulation* 2004;110:368–73.
7. Brouwer IA, Heeringa J, Geleijnse JM, Zock PL, Witteman JC. Intake of very long-chain n-3 fatty acids from fish and incidence of atrial fibrillation. The Rotterdam Study. *Am Heart J* 2006;151:857–62.

8. Calo L, Bianconi L, Colicicchi F, et al. N-3 Fatty acids for the prevention of atrial fibrillation after coronary artery bypass surgery: a randomized, controlled trial. *J Am Coll Cardiol* 2005;45:1723–8.
9. Anand RG, Alkadri M, Lavie CJ, Milani RV. The role of fish oil in arrhythmia prevention. *J Cardiopulm Rehabil* 2008;28:92–8.
10. Crystal E, Garfinkle MS, Conolly SS, Ginger TT, Sleik K, Yusuf SS. Interventions for preventing post-operative atrial fibrillation in patients undergoing heart surgery. *Cochrane Database Syst Rev* 2004; CD003611.
11. Pratt CM, Reiffel JA, Ellenbogen KA, Naccarelli GV, Kowey PR. Efficacy and safety of prescription omega-3-acid ethyl esters for the prevention of recurrent symptomatic atrial fibrillation: a prospective study. *Am Heart J* 2009;158:163–9.
12. Harris WS, Mozaffarin D, Lefevre M, et al. Towards establishing dietary reference intakes for eicosapentaenoic and docosahexaenoic acids. *J Nutr* 2009;139:804S–19S.
13. Mozaffarin D, Rimm EB. Fish intake, contaminants, and human health: evaluating the risks and the benefits. *JAMA* 2006;296:1885–99.
14. Harris WS. International recommendations for consumption of long-chain omega-3 fatty acids. *J Cardiovasc Med* 2007;8:Suppl 1:S50–2.

Do the Results of the German Pilot Phase of the EuroCMR Registry Indicate That the Chasm Has Been Crossed?

Geoffrey Moore (1) in his landmark book about the technology adoption lifecycle, “Crossing the Chasm,” divides the adopters into 5 phases—innovators, early adopters, the early majority, the late majority, and the laggards—with the curve forming a typical distribution curve with the major hump comprising the early and the late majority. Psychologically, others have classified these groups as technology enthusiasts (innovators), visionaries (early adopters), pragmatists (early majority), conservatives (late majority), and skeptics (laggards). Moore proposed that, for various reasons, there exists a chasm between the early adopters and the early majority and the success of any technological innovation depends on the capability of the technology to cross this chasm.

The results of the EuroCMR (European Cardiovascular Magnetic Resonance) Registry (2) seem to suggest that, at least in Germany, cardiac magnetic resonance imaging (MRI) has been able to cross this chasm with its value being understood and exploited for improving clinical care. Until now cardiac MRI has remained a niche exam with numerous cardiac MRI units performing studies for patients with congenital heart disease, cardiac masses, pericardial disease, aortic pathology, and angiography. The use of cardiac MRI in cardiomyopathies and ischemic heart disease has clearly changed the status of this technique in our diagnostic armamentarium. Because ischemic heart disease comprises the largest component of patients in contemporary clinical practice, the use of cardiac MRI in this arena for ischemia and viability assessment would seem to suggest that cardiac MRI has “crossed this chasm” and is now believed to be ready for primetime by cardiologists who do not perform cardiac MRI (because they would comprise the early and the late majority in Moore’s description).

To be sure this has happened, which is welcome. Given the potential of this technique, we wish to clarify the following points:

1. Of the referrals made, how many were made by physicians who actively perform cardiac MRI and by those who do not perform cardiac MRI? This information would eliminate the potential for a referral bias by those who perform this technique. In 23.1% of patients, cardiac MRI was the first technique ordered—who referred these patients? We consider any referral from any member of any cardiac MRI group to be a potential self-referral—this could simply mean that those who perform the technique might be able to decide which patient might be best suited for a cardiac MRI study.
2. In the subgroup of those who underwent cardiac MRI as the first study, all imaging needs were satisfied in 80.3%. Some information in the remaining 20% who required further studies would be helpful for understanding the limitations of cardiac MRI.
3. We are also interested to know the reasons why 11% of those who underwent viability assessment by cardiac MRI still required a subsequent transthoracic echocardiography, because cardiac MRI is now considered the practical gold standard for viability assessment. We acknowledge that the reasons might be other than viability assessment.
4. A total of 5,025 patients in the centers in the registry underwent cardiac MRI for suspected coronary artery disease/ischemia assessment and determination of viability. It would be very informative to know the number of single-photon emission computed tomography and echocardiographic studies ordered in these centers for the same reasons, because this would reflect the positioning of the cardiac MRI study in the routine clinical practice in the centers that participated in the registry.

The authors have made a laudable effort in extending the use of a very valuable technique. If the clarifications we request indicate that the chasm has not been crossed by cardiac MRI, we hypothesize that the presentation of prognostic data based on cardiac MRI studies might be the eventual trigger and the authors have accurately emphasized the need for extension of their work.

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REFERENCES

1. Moore GA. *Crossing the Chasm*. New York, NY: Capstone Publishing Limited, 1998.
2. Bruder O, Schneider S, Nothnagel D, et al. European Cardiovascular Magnetic Resonance (EuroCMR) registry: results of the German pilot phase. *J Am Coll Cardiol* 2009;54:1457–66.

Reply

We deeply appreciate Drs. Thomas and Tavares’ interest in our paper (1) and welcome their important points for further clarification.

We easily admit that some of us are enthusiasts (innovators), some are even visionaries (early adopters), but the majority are very pragmatic people who want cardiac magnetic resonance (CMR) to

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